1. Problem and cost

Contracts and forms must be typed into systems of record by hand.

A human review costs 1 per field. A wrong booked value costs 100.

Auto-booking pays only if accepted values are right more than 99% of the time (E = 100).

Break-even precision (E-1)/E: 99.0% for E = 100, 95% for E = 20, 99.8% for E = 500.

The dev set is mostly empty: 57.6% of the fields are null (absent).

Hallucinating a value for an absent field is the expensive mistake.

2. Pipeline and release rule

Full text, no truncation. Unreadable PDF -> null, no request. Chunks of 45k with 3k overlap, batches of 10 fields.

Prompt v3 sends the full field schema including all $defs. Schema validity and textual evidence are checked separately.

Each sample: a field is valid only if it passes both checks. Invalid values are never accepted.

probly: categorical distribution over the K valid answers, entropy in nats (diagnostic and method k_entropy).

Methods: k1_grounded, k_agreement_only, k_agreement_conf (min of share and confidence), k_entropy.

Freeze on dev_select, audit once on dev_audit. Active only if every release criterion holds.

Current decision: all-abstain (threshold 1.0).

3. Measured results (historical, prompt v2)

6 dev_cal docs, K=1, thr 0.95 | 199 fields | 2 acc, 1 wrong | net -49.2%

error cost 20: -9.0%, 500: -250.3%

dev_select, K=3, frozen 0.99 | 164 fields | 14 acc, 0 wrong | net +8.5%

precision 100%, coverage 18.9%

dev_audit, K=3, frozen 0.99 | 130 fields | 4 acc, 1 wrong | net -73.8%

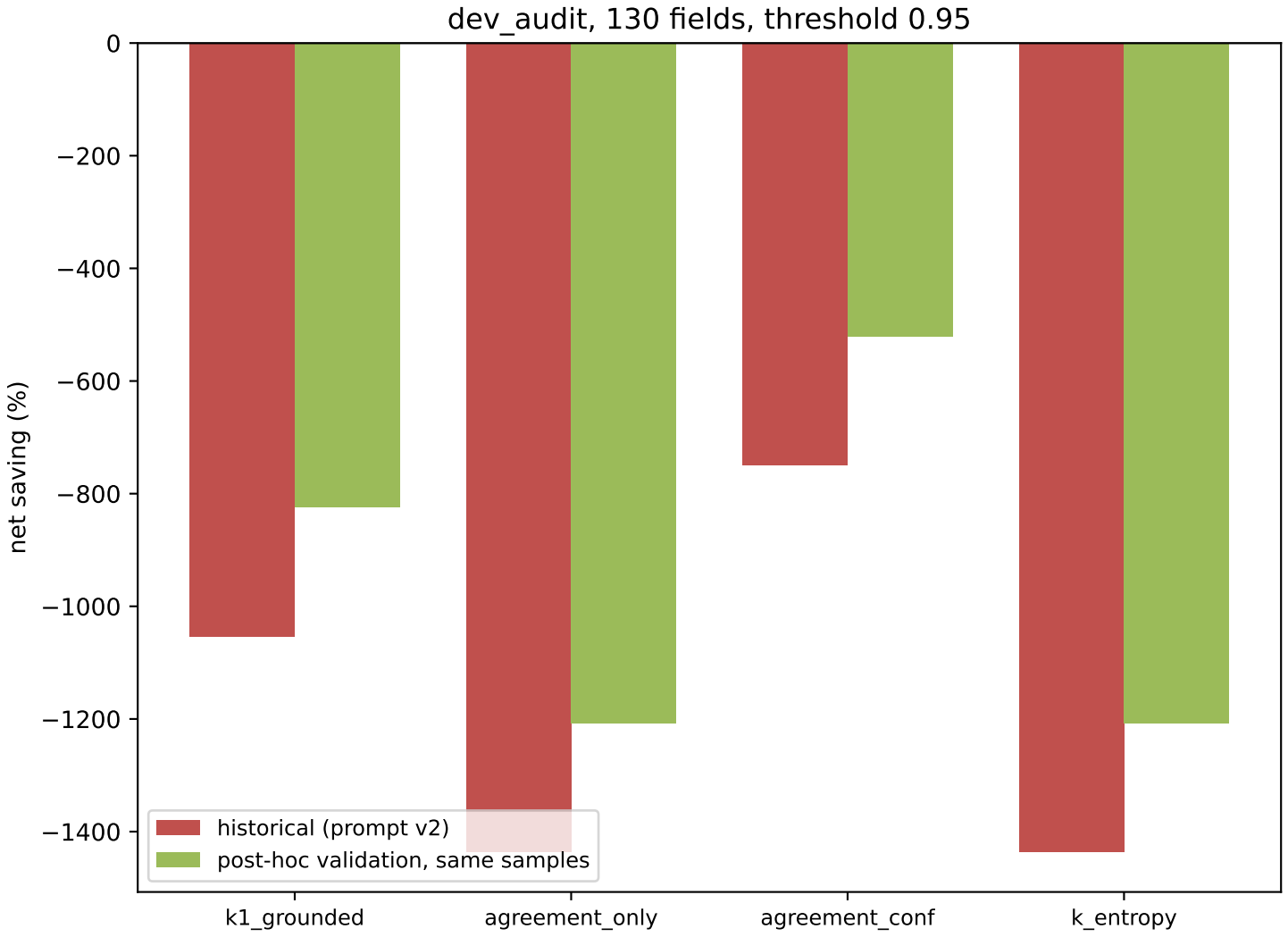
precision 75%, coverage 9.1%

Net saving = (accepted - 100 x (wrong + hallucinated)) / fields. Hallucinated values: 0 in all three sets.

The sets are very small (3 to 14 accepted fields). These are limited development measurements, not quality results.

Each K=3 set has 3 short CUAD contracts and 4 small ExtractBench documents (no scan).

4. Audit comparison at 0.95: historical vs post-hoc



Post-hoc counts (no new API call):

k1_grounded: 28 acc, 11 wrong, 0 hall
k_agreement_only: 31 acc, 15 wrong, 1 hall
k_agreement_conf: 22 acc, 7 wrong, 0 hall
k_entropy: 31 acc, 15 wrong, 1 hall

Every method is negative at 0.95 in both analyses.
The historical values were printed as factor 100 too small in

5. Error cause, limits and next measurement

Confirmed cause: the prompt omitted $defs, so the model invented row keys (labor_category, year_1) and currency text.

Second cause: nested values were checked only for text occurrence, not for keys or types. Fixed by schema validation.

Post-hoc: the wrong table value is no longer accepted. This is not a measurement of prompt v3.

Not done: prompt v3 on the API, Linux run, calibration (no predictor, no training data), test_public (not used).

Release rule: the 30-accepted-field minimum is a project heuristic, not a statistical result.

Next: a new K=3 measurement with prompt v3 on dev documents not used before, method and threshold frozen first.